

Agency MBS Weekly

In a New Hawk Minute

- **We maintain a neutral recommendation on the agency MBS basis.** While investors seem to have read the minutes -as modestly hawkish, the timing of balance sheet reduction, composition (MBS vs Treasuries) and the pace (tapering reinvestments versus ceasing them altogether) were very much in line with our expectations. FN 3.5s have outperformed their curve hedges by approximately 5-6 ticks since we turned neutral on March 23 and we continue to believe that near-term tailwinds from lower volatility, lower rates and relatively attractive valuations (versus corporates) could offset the longer-term risk from a tapering in Fed reinvestments.
- **Japanese investors may not buy till May.** Currency hedged dollar assets are looking attractively priced for Japanese investors and there is a looming question on whether they will ramp up their purchases in the fiscal new year that began on April 1. Contrary to common belief, historical data from the MoF and TIC suggest that foreign bond purchases in April were among the lowest of the year in the post-crisis period. Despite a low start in April, purchases tended to gradually gain traction starting in May. On average, we found purchases during the early part of the fiscal year (May-Nov) were higher than the later part (Dec-Mar).
- **We revise higher our 2017 net supply projection by \$30bn to \$215bn.** This projection continues to reflect our outlook for weaker housing activity year-over-year as data released since our last update supports our expectations on that front. The revision higher largely reflects the greater-than-expected spillover in Jan/Feb from late-2016 mortgage activity and banks' held-for-sale portfolios, as we discussed last month. Reperforming/modified pool supply from the GSE portfolios has also exceeded our expectation YTD and the new projection incorporates that. We also provide a summary of Q1 issuance, along with sector and servicer breakouts.
- **March Prepayment Commentary:** Aggregate Fannie 30yr fixed rate speeds increased 22% from 8.8 CPR to 10.8 CPR in March. The cohorts we track increased 24%, lower than our projection of up 28%. Speed increase was almost entirely driven by higher turnover which was likely up 35-40%, in line with our expectation. However, refi speeds were up less than the day count alone would suggest and surprised to the down side. We think excess capacity and declining loan volume may have weakened the dependence of refis on daycount.
- **Fed purchases to increase to \$23bn in April, up \$5bn from the previous cycle**
- **Gross Issuance \$93bn, net issuance estimated at \$15bn.** Net issuance decreased by roughly \$13bn MoM as paydowns increased.

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Stay Neutral

We maintain a neutral recommendation on the agency MBS basis. While investors seem to have read the minutes as modestly hawkish, the views expressed were very much in line with our expectations. FN 3.5s have outperformed their curve hedges by approximately 6-7 ticks since we turned neutral on March 23 and we continue to believe that near-term tailwinds from lower volatility, lower rates and relatively attractive valuations (versus corporates) could offset the longer-term risk from a tapering in Fed reinvestments.

Modestly Hawkish Minutes in line with our Expectations

The March FOMC minutes struck a slightly hawkish tone regarding the timing of balance sheet reductions. “Most participants” judged that “a change to the Committee’s reinvestment policy would likely be appropriate later this year”. While consistent with our view, this is earlier than market expectations for balance sheet reduction to be announced in 1H 2018.

“Participants generally preferred to phase out or cease reinvestments of both Treasury securities and agency MBS.” This is consistent with our base-line expectation that similar to tapering of asset purchases, the Fed will taper reinvestments of Treasuries and MBS together. The committee viewed ending reinvestments “all at once” as easier to communicate, but tapering as “reducing the risks of triggering financial market volatility.” We continue to believe that the second consideration will hold more sway, implying reinvestments are tapered – not ceased all at once.

While investors seem to have read the minutes as modestly hawkish, the timing of balance sheet reduction, composition (MBS vs Treasuries) and the pace (tapering reinvestments versus ceasing them) were very much in line with our expectations (see [US Economics & Multi-Asset Focus: Bringing down the Fed's balance sheet](#) for details). A summary of our views on Fed balance sheet expectations are as follows:

- **Timing:** Reinvestments will most likely be tapered in late 2017 or early 2018 when overnight rates are targeted between 1-1.5%.
- **Communication:** The FOMC statement will likely include language suggesting balance sheet normalization is forthcoming a few meetings ahead of the taper (Jun-Dec).
- **Composition:** Analogous to tapering of purchases, we expect an even Treasury/MBS tapering (\$5 billion each initially). We think the Fed will avoid increasing the proportional share of mortgages on the balance sheet. Since in some months projected MBS runoff is as low as \$15 billion, keeping the Tsy/MBS proportion would cap total (Tsy + MBS) runoff at \$30 billion per month.
- **Longer term runoff:** Once the balance sheet returns to its neutral level, Treasury securities may begin to increase. We think it is unlikely the Fed will allow MBS to run off the balance sheet at a pace much greater than \$15 billion per month. A faster pace may be more supply than the market can comfortably absorb.
- **Rates impact:** Winding down the Fed portfolio should increase term-premium and raise longer term yields. However, the initial taper may have a muted effect on longer-term interest rates as much of the increased Treasury issuance will be

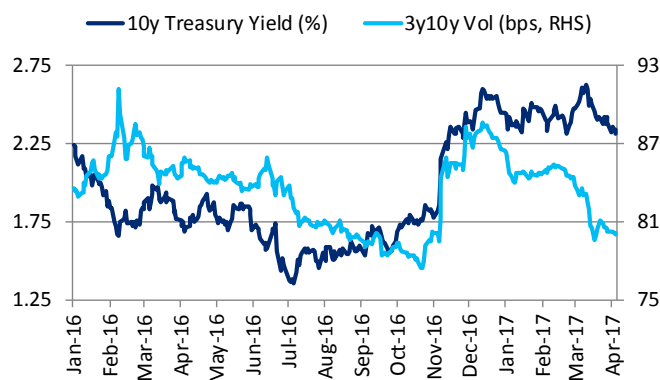
in the form of short-dated T-bills. Our base-case implies only about 5-15bp higher rates this year.

- **MBS impact:** We expect a 5-10bp widening in nominal MBS spreads as the Fed gives a clearer indication that they will start tapering reinvestments. Over the longer term, spreads could widen another 10-20bp. While market participants are dismissing the likelihood of active MBS sales, this will almost certainly occur at some point as otherwise some amount of MBS will remain on the balance sheet for as long as 30 years. Sales may come sooner if the Fed wishes to bring down the balance sheet faster but not increase the proportion of MBS.
- **Implications for banks:** Fed reserve assets on bank balance sheets will fall together with the Fed's balance sheet. But this reduction in supply of high quality liquid assets (HQLA) will be largely offset by increased supply of Treasuries and MBS to the private market. Also, much of the initial reserve reduction may occur relatively smoothly as foreign banks reduce reserve assets and deposit liabilities together.
- **Long term balance sheet size:** A sub-\$1 trillion balance sheet is no longer an appropriate steady state target for the Fed. While some former and current Fed officials support a larger balance sheet, there remains considerable disagreement regarding the long-term target. This is unlikely to delay the end of reinvestments but argues for a relatively slow pace of runoff. In our base case, the balance sheet falls to about \$2.6 trillion by 2023.

Near Term Tailwinds Remain

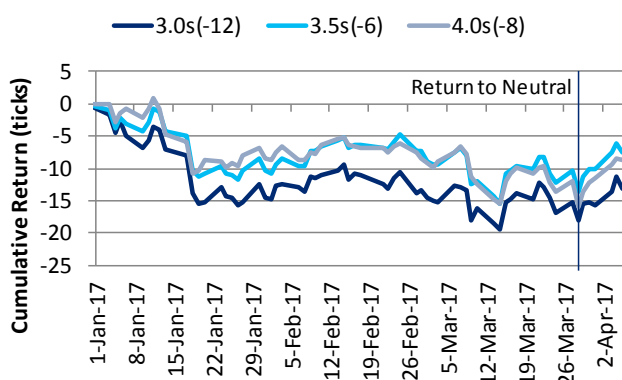
Production coupon MBS underperformed their curves hedges by 1-1+ ticks after the minutes were released, but outperformed hedges by 3 to 4 ticks over the week. 10-year Treasury yields rallied close to the lowest levels in 2017 (Figure 1) due to geopolitical risks, a weaker than expected NFP and the possibility of a delay in tax reform. Lower rates have pushed vega to 2017 lows as well, creating a positive environment for MBS.

Figure 1. 10y Treasury yields and Vega are both at their YTD lows



Source: Citi Research, Bloomberg

Figure 2. MBS have outperformed by 4 to 7 ticks since we moved from an underweight to neutral on March 23



Source: Citi Research, Yieldbook

FN 3.5s have outperformed their curve hedges by approximately 5-6 ticks since we turned neutral (Figure 2) on March 23 (see [Agency MBS Weekly: Repeal and Replace our Underweight](#) for details). We continue to believe that near-term tailwinds from lower volatility, lower rates and relatively attractive valuations (versus

corporates) could offset the longer-term risk from a tapering in Fed reinvestments. Carry on MBS has also improved as rolls have strengthened and lower rates have reduced hedge ratios. We maintain a neutral stance on the basis.

Summary and valuations

We present a summary of our views across the agency MBS sectors in Figure 3.

Figure 3. Summary of views across agency MBS sectors

	Summary View/ Risks	Reference Report
Basis	Maintain a neutral stance on the basis	Current report
Trade Recommendations	None	
Supply	Base case net supply forecast revised to \$215bn in 2017. Gross supply of \$1.23 trillion in 2017, 4% mortgage rate.	Current report
Demand	We reduce our bank demand estimate from \$150bn to \$100bn. We also reduce our overseas demand estimates from \$50bn to \$30bn, as Japanese investors are likely to prefer JGBs and Treasuries to MBS. Money managers and insurance companies will need to add \$50bn to balance supply/demand.	Agency MBS Weekly: Short and Sweet
Speed Projections	FN 30yr –Apr: -5% and G2s –Apr: -7%	Agency MBS Weekly: Repeal and Replace our Underweight
Coupon Stack	Neutral on the FN stack, prefer new issue 4.0s in G2s	Current report
G2/FN	Long term underweight G2/FN 3.0s	Current report
Specified Pools	Underweight specified pools vs TBA. We prefer 175k/200k max pools over LLB-HLB.	Agency MBS Weekly: Fed Fake
Short duration	Move to neutral from overweight 15s/30s but remain underweight ARMs vs 15s due to tight valuations. Quicken 5/1s offer best value in ARMs. Underweight 10s vs 15s on recent spread tightening. Neutral 20s v 30s. Within sub-3y bucket, floaters look attractive.	Agency MBS Weekly: Cash or Carry?

Source: Citi Research

Neutral on FN stack, prefer new issue 4.0s in G2s

Figure 4. WoW Performance in ticks
(Thursday to Thursday close).

Curve Hedged vs. Treasuries	FN	G2	DW
2.5s	5.1		2.1
3.0s	3.1	3.1	4.7
3.5s	4.2	1.8	-0.1
4.0s	4.2	5.2	
4.5s	3.4	2.1	
Duration Hedged Swaps	G2/FN	DW/FN	
2.5s		1.5	
3.0s	0.3	3.2	
3.5s	-1.9	-1.7	
4.0s	1.8		

Note: DW/FN swaps are vs +50bp FNCL
Source: Citi Research, Yieldbook

Figure 4 shows MBS performance against Treasury curve hedges over the last week. 30-yr conventionals outperformed by 3 to 4 ticks while G2s outperformed by 2 to 5 ticks. 15-yr PTs outperformed by 2 to 5 ticks as well.

The conventional 30-year stack continues to look fairly valued to us (Figure 5). The rally in rates should once again elevate prepay concerns on FN 4.0s but the coupon still outperformed by 4 ticks over the past week. We see very little relative value across the stack and recommend a neutral stance.

TBA 4.0s are looking attractively priced across the G2 stack but the coupon could underperform as the newly introduced 6 month seasoning rule across G2s may pressure higher coupon rolls. This may not occur till we see the Jan/Feb 4.0 pools spike in July/August. In the short-term, the roll and coupon could continue to do well as speeds remain slow. For long-term investors, we prefer owning the new issue pool at a payout.

G2/FN swaps were mixed over the past week (Figure 4). Although lower coupon swaps continue to look rich, we recommend moving to a neutral on G2/FN 3.5s as we expect Japanese demand to pick up in April. We maintain an underweight on G2/FN 3.0s.

Figure 5. Valuations on the 30-yr FNCL coupon stack

Coupon	WALA/WAC/ ALS	Price	Yield (%)	HAC (ticks)	zv- spread	OAS (bps)	Dur (yrs)	Conv	1y CPR
FN 3.0s	8/3.65/317	99-12+	3.07	1.3	63	19	6.2	-2.0	4
FN 3.5s	10/4.09/301	102-145	3.09	2.5	73	19	5.1	-2.5	7
FN 4.0s	20/4.59/272	105-006	2.92	3.4	79	21	3.8	-2.7	15
FN 4.5s	34/5.04/257	107-117	2.67	3.2	73	18	2.8	-2.3	21

Source: Citi Research, Yieldbook, March 30 2017 Closes

Figure 6. Valuations on the 30-yr G2SF coupon stack

Coupon	WALA/WAC/ ALS	Price	Yield (%)	HAC (ticks)	zv- spread	OAS (bps)	Dur (yrs)	Conv	1y CPR
G2 3.0s	7/3.40/276	101-026	2.83	1.1	46	-2	5.6	-2.2	6
G2 3.5s	9/3.87/249	103-235	2.81	2.3	55	4	4.5	-2.3	10
G2 4.0s	18/4.36/220	105-227	2.69	3.1	61	10	3.4	-2.2	18

Source: Citi Research, Yieldbook, March 30 2017 Closes

Figure 7. Spread and carry differential along with recommended positioning on G2/FN swaps

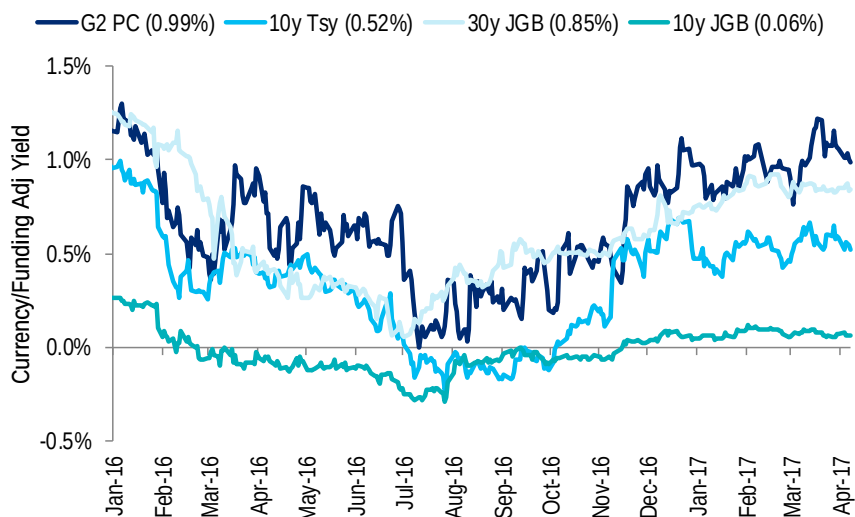
G2/FN	Yield	z-spread	OAS	HR	HAC	Recommended Positioning
3.0s	-0.24	-18	-21	92%	0.0	Underweight
3.5s	-0.29	-18	-15	91%	0.2	Neutral
4.0s	-0.23	-18	-11	90%	0.2	Neutral

Source: Citi Research, Yieldbook, March 30 2017 Closes

Japanese investors may not buy till May

Currency hedged dollar assets are looking attractively priced for Japanese investors (Figure 8) and there is a looming question on whether they will ramp up their purchases in the fiscal new year that began on April 1. Contrary to the common belief, historical data from the MoF and TIC suggest that foreign bond purchases in April were among the lowest of the year in the post-crisis period. Despite a low start in April, purchases tended to gradually gain traction starting in May. On average, we found purchases during the early part of the fiscal year (May-Nov) were higher than the later part (Dec-Mar).

Figure 8. Currency hedged yield on dollar assets has gone down, but remains attractive



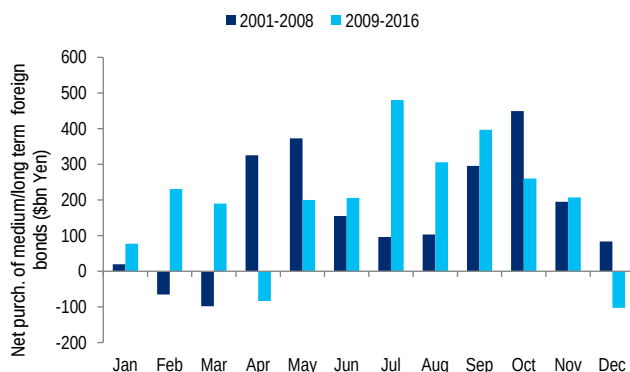
Source: Citi Research, Bloomberg

Weekly data from the Ministry of Finance

We look at two data sets to gauge the seasonal purchase pattern of Japanese investors – their weekly net purchases of foreign bonds from the Ministry of Finance and the monthly transactions volume from the TIC data.

Figure 9 shows the average monthly net purchase volumes of foreign securities by Japanese investors, broken down across the pre-crisis (2001-2008) and post-crisis (2009-2016) period. While Japanese investors did ramp up purchases of overseas bonds in April in the pre-crisis period, this has not really occurred more recently in the post-crisis period. In fact, in the 2009-2016 period, purchases in April have been amongst the lowest in the year. Overall, net purchases do not show much seasonality but purchases tend to be more concentrated in the early part of the fiscal year (May-Nov).

Figure 9. Average monthly net purchase of median/long term foreign bonds since 2009



Source: Citi Research, MoF

Figure 10. Each year, weeks with large net sales appear to be around the end/beginning of the fiscal year.

Year	Week with notable sales	Net sales (\$bn Yen)	Rank of net sales in the year
2008	28-Mar	(834)	4
2009	3-Apr	(2,108)	1
2010	7-May	(1,567)	1
2011	1-Apr	(1,178)	4
2012	6-Apr	(1,904)	1
2013	5-Apr	(1,140)	4
2014	28-Mar	(764)	6
2015	3-Apr	(3,072)	1
2016	1-Apr	(1,556)	1

Note: Dates represent the end of the week
Source: Citi Research, MoF

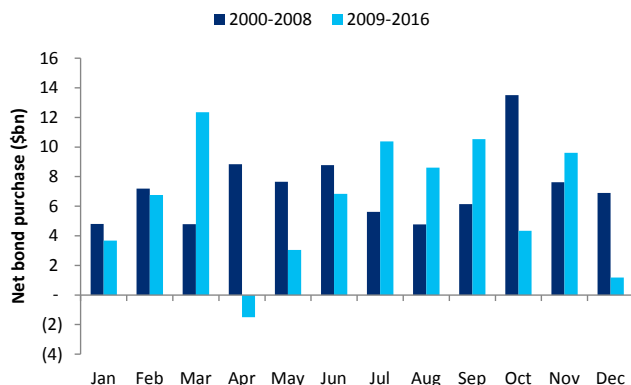
Notably, there is a trend of selling foreign securities at the end/beginning of the fiscal year (late Mar to early Apr, Figure 10). For instance, in 2009, 2015 and 2016, the largest weekly net sales of foreign bonds took place during the week ending in early April. In other years such as 2012, the largest weekly sale was in the first week of April. Net sales were also quite large in late March of other years. The selling of foreign securities at the end of the fiscal year may be related to their repatriation of foreign assets back to Japan, though selling in the beginning of the fiscal year is somewhat surprising.

Treasury International Capital (TIC) data points to similar trends

The monthly TIC data, which only includes Japanese purchases of dollar securities, provides us with another way of checking the seasonality in purchases. Similar to the MoF data in Figure 10, Figure 11 presents monthly net purchases broken down across the pre-crisis and post-crisis periods and confirms that trend observed in the MoF data. April purchases were high in the pre-crisis period but the lowest in the post-crisis period, and purchases in May-Nov were generally more robust than Dec-Mar in their fiscal year.

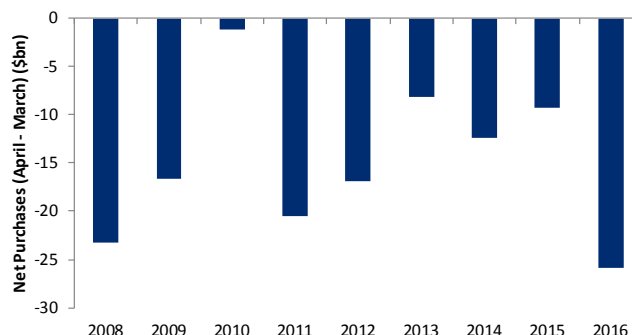
Looking at average monthly volume could be misleading due to much lower overall volumes in early years. Another way to look at whether Japanese investors ramp up purchases in April is how net buying/selling in April compares with March each year (Figure 12). It shows that Japanese investors purchased fewer bonds in April vis-à-vis March every year since 2008. Investors who expect heavy purchases from Japan in April may have to wait more patiently.

Figure 11. Japanese investors ramped up their purchases of dollar bonds in April pre-crisis, while the post crisis period shows April purchases were the lowest on average



Note: Data include Treasury, agency deb, MBS and corporate bonds
Source: Citi Research, TIC

Figure 12. Net buying in April has been much lower than net buying in March in the post crisis period



Source: Citi Research, TIC

Revising MBS net supply +\$30bn to \$215bn

We revise higher our 2017 net supply projection by \$30bn to \$215bn. This projection continues to reflect our outlook for weaker housing activity year-over-year as data released since our last update supports our expectations on that front. The revision higher largely reflects the greater-than-expected spillover in Jan/Feb from late-2016 mortgage activity and banks' held-for-sale portfolios, as we discussed last month. Repurchasing/modified pool supply from the GSE portfolios has also exceeded our expectation YTD and the new projection incorporates that.

Revising projections on supply spillover and mod issuance

Figure 13. 2017 Gross and Net MBS Supply Projections Revised to \$1.23tn and \$215bn at 4.0% Mortgage Rate

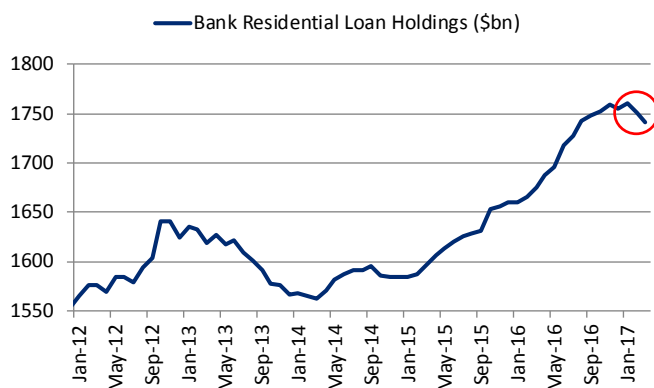
	2011	2012	2013	2014	2015	2016	2017 Proj. 4.0% Mtg. Rt.	2017 +50bps	2017 -50bps
Gross Agency MBS Supply	1142	1661	1546	925	1,251	1,472	1230	1030	1530
Net Agency MBS Supply	29	33	235	71	164	230	215	170	275
DRIVERS OF NET SUPPLY:									
Home purchases, Non-agency refis, and Cashouts add to MBS net supply									
New home sales	29	40	52	55	69	84	84	79	92
Non-agency MBS voluntary prepaes	76	65	73	54	61	59	42	35	53
Cashouts and loan consolidations	54	65	59	42	66	84	75	65	85
Scheduled Amortization reduces net supply									
Scheduled amortization of MBS universe	148	155	164	171	179	185	189	186	192
An increase in banks and GSE portfolios reduces net supply and vice versa									
Change in financial institution single-Family loan holdings	10	30	-49	23	70	95	55	50	70
Change in GSE loan portfolios	-10	-59	-97	-46	-51	-47	-60	-60	-60
Remaining Net supply not explained by above factors (primarily from Existing home sales not captured in other categories)	18	-11	69	68	167	235	198	167	247

Source: Citi Research, FDIC, SNL, 1010data, Fannie Mae, Freddie Mac, Bloomberg

Figure 13 shows our updated net supply projections under various rate scenarios. These projections reflect the \$25bn mark-to-market risk that we had acknowledged

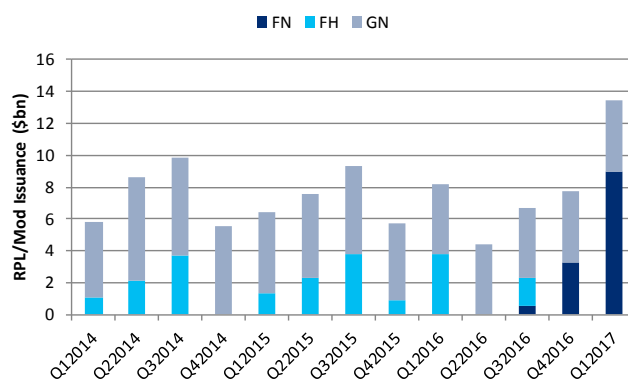
last month ([Agency MBS Weekly, March 3, 2017](#)). Bank loan holdings have declined by ~\$20bn YTD (Figure 14), lending support to our thesis that there was a flush of supply from bank balance sheets. Seasoned supply was in line with our initial projections, suggesting that much of this decline likely came out of the held-for-sale portfolio, as anticipated, although this can be confirmed only when the detailed data on bank portfolios is available next month. Reperforming/modified pool supply from the GSE portfolios (Figure 15) has exceeded the run-rate implied in our projection of \$20bn for the year and we incorporate that in our new forecasts as well.

Figure 14. Bank residential loan holdings have declined by ~\$20bn in 2017



Source: Citi Research, H.8.

Figure 15. 2017Q1 Mod/RPL issuance exceeded our expectations as Fannie Mae securitized \$9bn

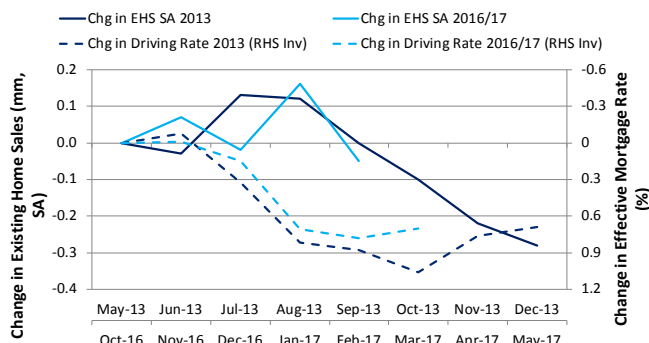


Source: Citi Research, 1010data

Data finally reflects a slowdown in housing

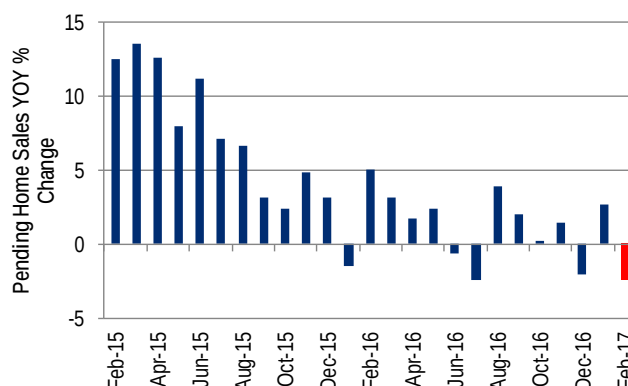
Since January, our projections have priced in a modest decline in home sales from last year, the effect of which we expected to be offset by higher loan size YoY ([MBS Supply Roundup and Projections](#)). After showing signs of persistent strength initially, existing home sales finally declined in February (Figure 16). This exhibited a lagged reaction to the rate selloff, similar to that seen after the taper tantrum in 2013, a possibility that we had raised last month. Moreover, pending home sales for February were lower than year-ago levels (Figure 17), suggesting potential for existing home sales to lag further in March. Turnover speeds for March also suggest that housing activity would be down in March YoY, adjusted for day count difference, as we discuss in the prepayment commentary below. We continue to monitor the housing data closely for evolving trends in the housing outlook.

Figure 16. Existing home sales declined in February exhibiting a similar lagged response to the Taper Tantrum of 2013



Source: Citi Research, Bloomberg

Figure 17. Pending home sales in February were 2.4% lower than Feb 2016



Source: Citi Research, NAR

Issuance by product – Q1 net issuance at \$99bn

Gross issuance in Q1 came in at \$323bn, \$116 lower than from Q4 last year but \$56bn higher compared to Q1'16. Q-o-Q gross issuance declined in all products besides conventional ARMs. Net issuance in Q1 reached \$99bn, similar to Q4 last year but \$73bn higher than a year ago. GN net issuance accounted for roughly one-third of total net issuance, similar to last quarter but much lower the Q1-Q3 of 2016.

Figure 18. Quarterly gross issuance

Quarter	Conventional					Ginnie					Conventional+ Ginnie				
	30s	15s	ARM	Others	Total	30s	15s	ARM	Others	Total	30s	15s	ARM	Others	Total
Q1'16	126	26	4	17	173	80	2	1	10	94	206	28	5	27	267
Q2'16	163	32	4	19	218	104	2	1	16	123	267	35	5	35	341
Q3'16	208	41	4	30	283	119	3	1	20	143	327	44	5	49	426
Q4'16	210	49	3	38	300	120	3	1	15	139	330	52	4	53	439
Q1'17	150	33	5	30	218	90	2	1	11	104	241	35	6	41	323
Q1'17-Q4'16	(59)	(16)	2	(7)	(82)	(30)	(1)	(0)	(4)	(34)	(89)	(17)	1	(12)	(116)
Q1'17-Q1'16	24	6	1	14	45	11	0	(1)	1	11	35	7	0	14	56

Source: Citi Research, 1010data

Figure 19. Quarterly net issuance

Quarter	Conventional					Ginnie					Conventional+ Ginnie				
	30s	15s	ARM	Others	Total	30s	15s	ARM	Others	Total	30s	15s	ARM	Others	Total
Q1'16	22	(4)	(6)	(5)	7	15	(0)	(1)	5	19	37	(5)	(7)	1	26
Q2'16	25	(3)	(9)	(7)	6	21	(0)	(2)	9	28	46	(4)	(10)	2	33
Q3'16	41	2	(8)	(1)	35	25	(0)	(2)	12	34	66	2	(10)	11	69
Q4'16	53	12	(8)	9	65	31	0	(2)	7	37	84	12	(10)	16	102
Q1'17	55	2	(5)	10	62	33	0	(2)	6	37	88	2	(7)	16	99
Q1'17-Q4'16	2	(10)	3	1	(3)	1	(0)	0	(1)	0	3	(10)	4	(0)	(3)
Q1'17-Q1'16	33	7	1	15	55	17	0	(0)	1	18	50	7	0	15	73

Source: Citi Research, 1010data

Servicer rank in Q1– Wells, Chase top conventional list

Figure 20 lists 30yr issuance characteristics of the top 10 servicers in Q1 for each agency. We discuss some key highlights below.

- **Wells and Chase top the conventional list.** Wells remains the largest issuer for conventional 30yr while ceding the top spot in Ginnies to PennyMac. Its share in conventional 30yr remained high at 18-19% while its share in Ginnies dipped below 10%. Chase moved to the second largest conventional 30yr issuer, followed by Quicken. It's possible that the increase in Chase issuance was driven by its lower loan retention.
- **Banks rely heavily on correspondent channel.** The correspondent loans of Wells/Chase/US Bank account for over 50% of their issuance, much higher than the 35-40% on average. However, most bank servicers largely moved away from the broker channel.
- **Overall, top originators have a much higher TPO share than average,** particularly in Ginnies where PennyMac, Amerihome, and PacUnion all have over 80% TPO share. In aggregate, GN's TPO share continues to outpace conventionals.
- **Quicken has the highest cashout share.** Quicken's share of cashout loans is one of the highest in both Freddie and Ginnie, though it's partially due to its high total refi issuance. At the other end of the spectrum in Ginnies, servicers that have the lowest cashout share of refi are Wells, Chase, Freedom and PacUnion.

Figure 20. Gold, FN and GN 30yr issuance by servicers

	UPB	WAC	FICO	Avg Insize (\$k)	LTV	Refi%	Broker%	Corr%	Cashout%			
Freddie												
All	66.1	4.11	747	241	76	48%	10%	38%	23%			
Wells	12.7	4.09	744	249	76	45%	0%	68%	22%			
Chase	5.5	3.95	754	252	73	55%	0%	61%	28%			
US Bank	3.4	3.98	750	230	78	46%	7%	64%	18%			
Quicken	3.3	4.25	736	212	76	69%	10%	1%	35%			
Caliber	2.9	4.31	746	261	78	29%	33%	24%	15%			
Amerihome	2.6	4.2	747	272	80	36%	5%	94%	17%			
LoanDepot	1.6	3.9	753	220	78	42%	0%	75%	18%			
BBT	1.6	4.41	748	265	77	41%	90%	10%	25%			
United Shore	1.4	3.95	754	216	70	58%	0%	0%	27%			
BoA	1.1	4.27	737	239	71	79%	0%	24%	28%			
Fannie												
All	82.9	4.15	742	225	76	47%	10%	34%				
Wells	14.7	4.00	748	253	76	43%	0%	66%				
Quicken	4.7	4.27	731	215	72	78%	7%	0%				
Chase	4.0	3.9	755	231	73	55%	0%	52%				
Suntrust	2.5	3.94	751	237	74	51%	0%	67%				
Pennymac	1.8	4.29	750	279	81	29%	6%	94%				
USBank	1.6	4.38	731	178	85	22%	5%	78%				
Pennymac	1.4	4.23	750	236	73	41%	7%	93%				
Flagstar	1.3	4.18	741	246	73	53%	20%	71%				
United Shore	1.3	4.38	747	249	77	45%	85%	14%				
Freedom	1.3	3.97	752	252	75	70%	7%	19%				
Ginnie												
	UPB	WAC	FICO	Avg Insize (\$k)	LTV	Refi%	Broker%	Corr%	Cashout%	Mod%	FHA%	VA%
All	89.3	3.83	691	216	93	37%	13%	42%	15%	5%	57%	39%
Pennymac	9.3	3.81	700	231	94	30%	15%	71%	14%	3%	56%	39%
Wells	7.9	3.73	716	219	94	24%	1%	60%	12%	19%	46%	51%
Freedom	7.3	3.78	679	204	94	55%	9%	48%	9%	2%	51%	37%
Quicken	4.2	3.82	692	192	86	73%	7%	0%	41%	1%	59%	41%
Caliber	2.9	3.84	687	251	94	25%	33%	40%	14%	1%	63%	34%
Amerihome	2.7	3.84	700	250	94	23%	4%	95%	13%	0%	73%	25%
US Bank	2.4	3.94	697	185	95	16%	0%	79%	42%	8%	73%	22%
LoanDepot	2.4	3.78	689	245	91	70%	22%	0%	9%	1%	51%	49%
Chase	2.3	3.69	681	208	94	19%	0%	74%	15%	16%	45%	47%
PacUnion	2.1	3.83	669	223	93	43%	13%	70%	11%	1%	60%	33%

Source: Citi Research, 1010data

March Prepayment Commentary

Figure 21. Driving factors for prepayments

	Feb	Mar	MoM%
PMMS rate	4.20	4.17	-0.03
MBA refi Index	1125	1176	5%
Average loan size	245	251	3%
Day Count	19	23	21%
Turnover seasonal	0.76	0.9	18%

Source: Citi Research

Aggregate Fannie 30yr fixed rate speeds increased 22% from 8.8 CPR to 10.8 CPR in March. The cohorts we track increased 24%, lower than our projection of up 28%. FN 15s were up 25%, more than 30s. Gold speeds increased less than Fannies' with 30s and 15s speeds up 19% and 16%, respectively. G2 30s increased by 15%, less than conventionals.

FN 3.0s led the pack across the coupon stack, speeding up 43% in aggregate (Figure 22, Figure 23). However, Gold 3.0s were up only 34%, considerably less than FNs. FN 3.5s and 4.0s were up 24% and 18% respectively, in line with Golds. Further up the coupon stack, 4.5s to 5.5s were up 11-14%, less than the 21% increase in day count.

Figure 22. Fannie 30yr speeds by coupon

Coupon	Feb	Mar	MoM%
2.5	3.4	5.3	54%
3.0	4.4	6.3	43%
3.5	7.4	9.2	24%
4.0	11.1	13.1	18%
4.5	14.4	16.0	11%
5.0	16.9	19.3	14%
5.5	17.9	20.5	14%

Source: Citi Research, eMBS

Figure 23. Gold 30yr speeds by coupon

Coupon	Feb	Mar	Change
2.5	2.6	3.8	47%
3.0	4.3	5.8	34%
3.5	7.1	8.8	24%
4.0	11.1	12.8	16%
4.5	14.8	16.5	11%
5.0	17.9	19.3	8%
5.5	19.2	20.5	7%

Source: Citi Research, eMBS

Figure 24. FNCL cohort speeds

Cohort	Feb	Mar	% change	Act-Proj
3.0 2016	2.7	4.0	50%	0.1
3.0 2015	5.0	7.5	51%	0.9
3.0 2013	6.0	8.4	40%	0.3
3.0 2012	5.8	7.9	37%	0.0
3.5 2016	5.0	6.4	28%	-0.7
3.5 2015	7.5	9.7	30%	0.0
3.5 2014	9.5	11.7	23%	-0.3
3.5 2013	8.7	11.1	28%	-0.1
3.5 2012	8.5	10.2	21%	-0.9
4.0 2016	7.2	9.7	35%	-0.4
4.0 2015	11.1	13.9	25%	0.2
4.0 2014	13.1	15.9	21%	-0.3
4.0 2013	12.6	14.9	18%	-0.9
4.0 2012	11.1	13.5	21%	-0.7
4.0 2011	11.0	13.1	20%	-1.0
4.5 2014	15.3	18.6	21%	-0.1
4.5 2013	16.5	17.4	5%	-2.7
4.5 2011	14.0	15.0	7%	-2.5
4.5 2010	14.6	16.4	12%	-1.9

Source: Citi Research, eMBS

At the cohort level, the 2015-16 vintages increased more than the seasoned vintages, partly driven by the seasoning ramp. FN 3.0s speeds were largely in line with expectations, driven by an increase in turnover seasonals and day count. Seasoned cuspy and high coupons surprised to the downside as refs were much weaker than our projections. For instance, 2010-11 and 2013 4.5s were only up 5-12%, much less than our projection of up 20-25%.

The trends across the stack suggest that the speed increase was almost entirely driven by higher turnover while refi speeds were only slightly higher despite a 21% increase in day count and an 8% increase on the index. We think excess capacity and declining loan volume may have weakened the dependence of refs on daycount as discussed in detail below.

Turnover rebounded 38%, in line with expectation

Aggregate 30yr 2.5s and 15yr 2.0s speeds point to a 38% increase of turnover in March (Figure 25). Pending home sales also suggest that EHS in March could have reached 431k, a 37% increase from February (Figure 26). This is in line with the increase in daycount (+21%) and seasonals (+16%).

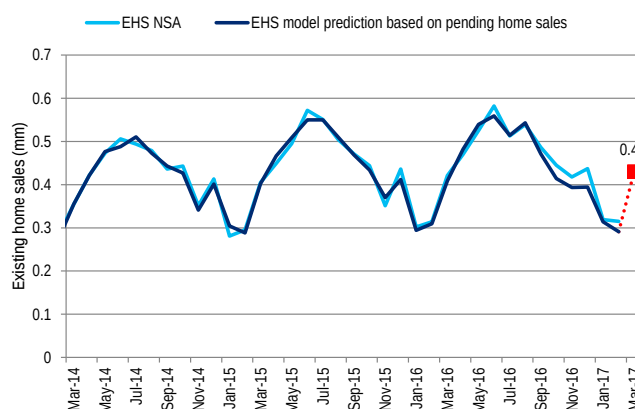
If March EHS is in line with our model projection, total home sales in Q1 would have tracked last year's volume fairly closely (Figure 27). However, daycount in March is one day more this year than 2016 suggesting daily average home sales could still be down slightly from last year, a sign of potential weakness driven by high rates. Whether this trend will hold through the year is unclear at this point and we await more data to confirm.

Figure 25. Aggregate 30yr 2.5s and 15yr 2.0s speeds suggest turnover could be up as much 38%

	Coupon	Bal (\$bn)	Feb CPR	Mar CPR	MoM%
FN 30yr	2.5	3.9	4.5	6.8	51%
	2.5	5.1	3.9	6.8	75%
Gold 30yr	2.5	0.9	4.3	5.0	18%
	2.5	1.2	4.2	6.9	64%
Total 30yr	2.5	11.2	4.2	6.7	60%
DW	2.0	10.5	5.9	8.0	36%
	2.0	3.3	6.3	8.2	29%
NG	2.0	6.6	6.3	7.3	16%
	2.0	1.4	6.8	7.8	16%
Total 15yr	2.0	21.8	6.1	7.8	27%
30yr+15yr	2.5/2.0	33.0			38%

Source: Citi Research, eMBS

Figure 26. Pending home sales suggest March EHS could reach 0.43mm, a 37% increase from February



Source: Citi Research, NAR

Figure 27. Daily average home sales in March could still be down slightly from last year even if EHS rise over 37% from February to 431k in March

	2016 EHS ('000)	Daycount	Daily EHS ('000)	2017 EHS ('000)	Daycount	Daily EHS ('000) Day	YoY Change of Daily EHS
Jan	302	19	15.9	319	20	16.0	0%
Feb	314	20	15.7	315	19	16.6	6%
Mar	421	22	19.1	431*	23	18.7	-2%
Q1 Total	1037	61	17.0	1065	62	17.2	1%

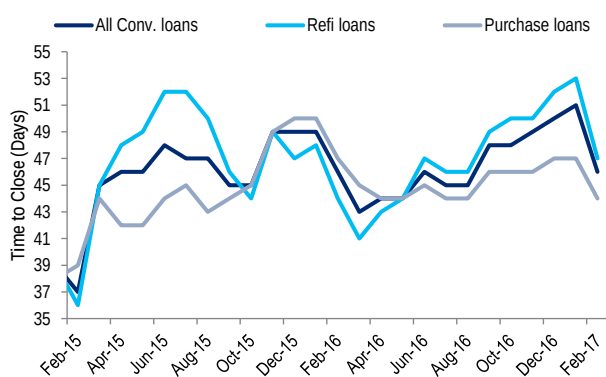
*Note: Mar'17 EHS is projected based on pending home sales and daycount, as shown in Figure 26 above.
Source: Citi Research, NAR

Low refis came in as a surprise

Given a slightly higher index and 21% increase in daycount, refi speeds remained surprising low and appeared to be less impacted by daycount this month. To be sure, given the ample capacity in the industry, the number of applications being process per day may decline, which may dilute the impact of daycount on prepayments. In our projections, we had dialed down the impact of daycount on refis, though the actual print came out even lower than our expectation.

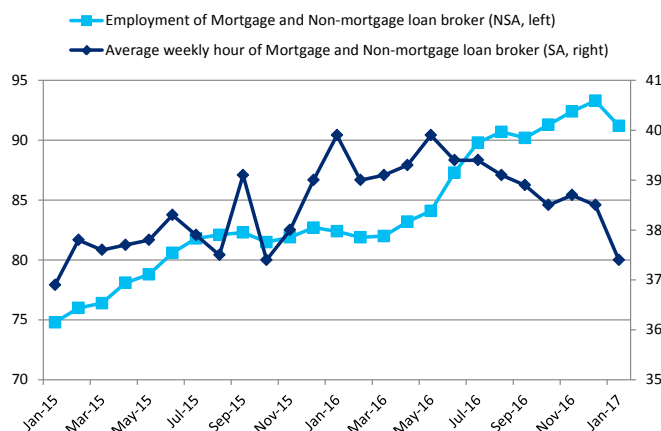
Based on the Ellie Mae origination report, backlog of loans in the pipeline were largely flushed out in January, as closing time was back to a more normal level in February after rising for the previous several months (Figure 28). The BLS data also shows that employment of loan brokers dipped slightly in January while the average weekly hours dropped to the lowest point in the last couple of years (Figure 29). These trends suggest that the number of loans being processed per day has likely changed in the last two months, and this may have dampened the effect of daycount on refis.

Figure 28. Loan closing time was back to more normal level in February



Source: Citi Research, Ellie Mae

Figure 29. Employment and average weekly hours for loan brokers dipped in January



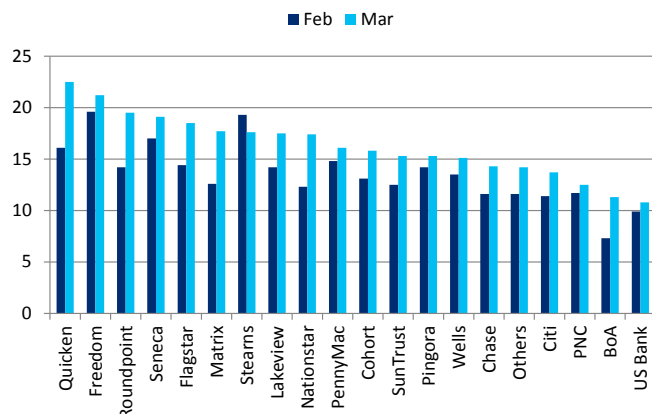
Source: Citi Research, BLS

Servicer speeds – non-bank servicer speeds increase in 3.0s

Despite overall low refi speeds, faster servicers managed to outpace cohort speeds by quite a margin. While the speed changes across servicer for 2013 4.0s seem broadly in line, it is worth highlighting that across 2013 3.0s, Pingora, Quicken and Freedom paid north of 10 CPR, 2-3 CPR faster than the cohort. More importantly,

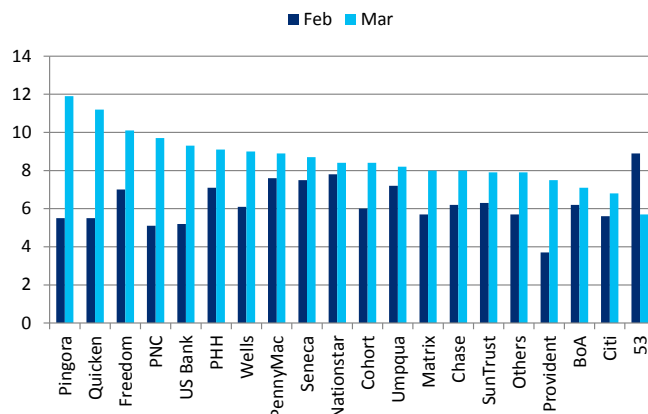
speeds almost doubled for 2013 FN 3.0s serviced by Pingora and Quicken versus the 40% increase in the cohort. Some of these borrowers may be cashing out or refinancing into shorter maturity products.

Figure 30. Servicer speeds in 2014 FN 4.0s



Source: Citi Research, 1010data

Figure 31. Servicer speeds in 2013 FN 3.0s

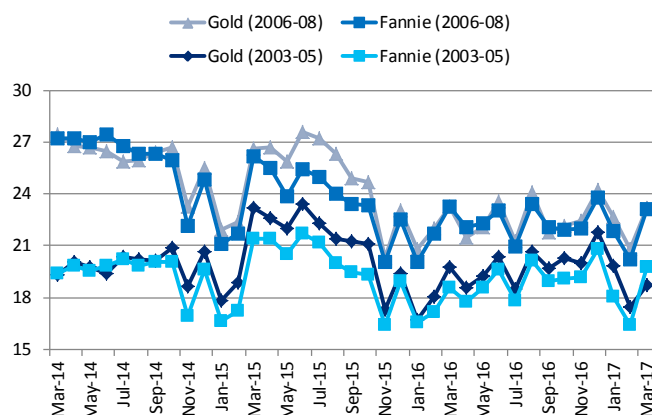


Source: Citi Research, 1010data

HARP speeds up 16% in March; large jump in cash window speeds

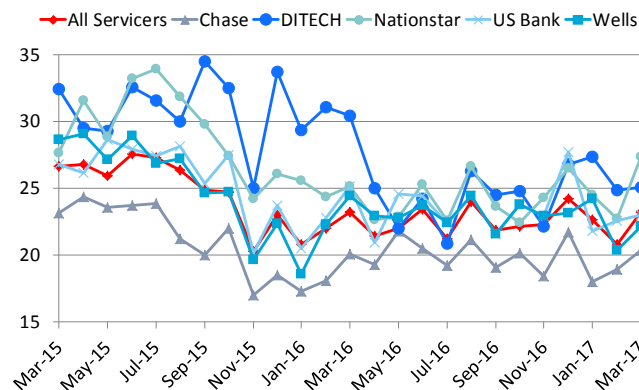
Aggregate HARP speeds were up just 16% in March, as much lower refi activity offset the speed increase from turnover seasonal effects and higher daycount. After three months of resiliently high speeds, it appears that the increased solicitation of these borrowers following the elections may have run its course. Despite the relatively high speeds following the election, conventional pre-HARP loans are paying in line year-over-year.

Figure 32. HARP loans saw a smaller increase in prepayment than post-HARP loans (2003-05 and 2006-08 vintages)



Source: Citi Research, 1010data

Figure 33. Speeds were mixed across bank servicers but slowed modestly for non-bank servicers (2006-08 Gold 5s-6.5s)



Source: Citi Research, 1010data

As we saw in post-HARP collateral, there was a distinctive difference between the speeds in Fannie and Freddie pools (Figure 32). Fannie pre-HARP pools sped up by 18% while Freddie increased by just 9%. The 2003-2005 Fannie vintages increased by over 20% on aggregate, led by a 71% jump in 2004 FN 6.0s. This large jump was primarily caused by recently issued Fannie cash window pools

(such as AS9272 and AS9276) and if we exclude cash window pools from the calculation, 2004 FN 6.0s only increased by 21% (Figure 34).

Figure 34. Speeds of 2004 6.0s only increased 21% if we exclude cash window pools

	Feb CPR	Mar CPR	MoM % Chg
Cash Window*	15.2	46.8	209%
Non-Cash Window	15.3	18.5	21%
Aggregate	15.3	26.0	70%

*Note: Cash Window defined as pools with prefix AS or AB

Source: Citi Research, 1010data

Across servicers, Nationstar, the largest servicer of pre-HARP Freddie 2006-2008 loans, led the increase in speeds with a 20% jump (Figure 33). The large bank servicers saw a more modest increase with Wells, Chase and BoA increasing by 9%, 7% and 7%, respectively.

Specified Pools

Figure 35 and Figure 36 list selected spec pool speeds and their comparison with respective cohorts. Within specified pools, loan balance pools saw large increases in prepayments. In many cohorts, LLB-HLB pools saw greater percentage increases than their corresponding cohorts as the elevated turnover impacts their speeds more. More refi driven stories such as purchase-HiLTV saw smaller increases in speeds.

- **NY pools low turnover leads to smaller prepayment increases** (Figure 37). Across most cohorts, NY pools saw smaller increases than their respective cohorts or loan balance pools. FN 4.0s of 2014 NY pools, for example, increased by 18% compared to the cohort's 21% and LLB's 36%. NY loans generally have lower turnover than other loans so were less impacted by increase in turnover.
- **LowFICO and HiLTV narrow gap with 200k max in 2014 4s** (Figure 38). As refis remained slow in March, LowFICO and HiLTV pools generally saw smaller increases in speed than loan balance pools. While 2014 200k max FN 4.0s are still paying slower than corresponding LowFICO or HiLTV, the gap has narrowed. Should rates stay in this range, we could continue to see this gap narrow as refis become a smaller component of speeds.

Figure 35. Speeds of loan balance and CQ/CR spec pools

Coupon	Vintage	Spec	Bal(\$bn)	Jan	Feb	MoM	% of cohort
3.0	2013	Cohort	144.2	6.0	8.4	40%	
3.0	2013	LLB	1.4	6.1	9.4	54%	1.11
3.0	2013	MLB	2.7	7.3	9.0	23%	1.06
3.0	2013	HLB	6.1	6.8	9.7	43%	1.15
3.0	2013	HHLB	4.1	6.4	8.2	29%	0.98
3.5	2015	Cohort	167.6	7.5	9.7	30%	
3.5	2015	LLB	2.0	6.6	8.6	31%	0.89
3.5	2015	MLB	3.9	6.8	8.6	28%	0.89
3.5	2015	HLB	10.8	6.4	9.1	43%	0.94
3.5	2015	HHLB	7.9	6.5	8.9	36%	0.91
4.0	2014	Cohort	82.0	13.1	15.9	21%	
4.0	2014	LLB	2.6	8.8	11.9	36%	0.75
4.0	2014	MLB	4.2	9.0	11.2	23%	0.70
4.0	2014	HLB	9.1	10.3	13.4	30%	0.84
4.0	2014	HHLB	5.2	11.9	13.7	15%	0.87
4.0	2014	200k Max	0.6	12.2	16.5	35%	1.04
4.0	2015	Cohort	49.6	11.1	13.9	25%	
4.0	2015	LLB	2.4	7.9	9.5	19%	0.68
4.0	2015	MLB	3.7	7.8	11.1	43%	0.80
4.0	2015	HLB	7.9	9.1	10.8	19%	0.78
4.0	2015	HHLB	4.3	9.3	11.7	25%	0.84
4.0	2015	200k Max	2.3	8.4	13.3	59%	0.95
3.5	2012	CQ	7.6	8.3	10.6	28%	1.04
3.5	2012	CR	8.2	8.2	10.3	25%	1.00
3.5	2013	CQ	7.2	8.5	9.8	16%	0.89
3.5	2013	CR	7.9	7.1	8.6	21%	0.78
4.0	2012	CQ	6.0	10.0	14.2	42%	1.05
4.0	2012	CR	6.5	9.8	12.5	28%	0.93
4.0	2015	CQ	0.7	6.3	10.2	63%	0.74
4.0	2015	CR	0.4	2.9	6.7	134%	0.48

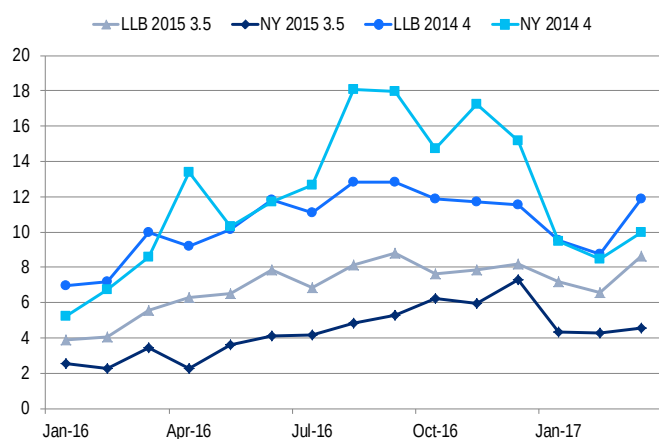
Source: Citi Research, 1010data

Figure 36. Speeds of other spec pools

Coupon	Vintage	Spec	Bal(\$bn)	Jan	Feb	MoM	% of cohort
3.0	2013	Cohort	144.2	6.0	8.4	40%	144.2
3.0	2013	HILTV	0.3	10.8	6.8	-37%	0.3
3.0	2013	LOWFICO	1.0	9.4	10.4	11%	1.0
3.0	2013	NY	1.0	3.4	6.4	85%	1.0
3.0	2013	Investor	1.0	7.2	9.0	25%	1.0
3.0	2013	CK	8.9	5.8	6.6	14%	8.9
3.5	2013	Cohort	72.2	8.7	11.1	28%	72.2
3.5	2013	NY	1.1	7.3	6.5	-11%	1.1
3.5	2013	HILTV	0.3	11.5	21.7	89%	0.3
3.5	2013	LOWFICO	3.7	9.0	14.6	62%	3.7
3.5	2013	Investor	3.0	8.0	8.5	6%	3.0
3.5	2013	CK	1.6	8.0	10.7	33%	1.6
3.5	2015	Cohort	167.6	7.5	9.7	30%	167.6
3.5	2015	NY	2.8	4.3	4.6	6%	2.8
3.5	2015	HILTV	4.5	7.5	9.6	28%	4.5
3.5	2015	LOWFICO	6.3	9.1	13.2	45%	6.3
3.5	2015	CK	3.4	7.8	10.1	29%	3.4
4.0	2014	Cohort	82.0	13.1	15.9	21%	82.0
4.0	2014	NY	1.6	8.4	10.0	18%	1.6
4.0	2014	HILTV	0.7	15.7	19.5	24%	0.7
4.0	2014	LOWFICO	4.9	15.9	18.0	13%	4.9
4.0	2014	Investor	1.1	12.7	16.3	28%	1.1
4.0	2014	CK	0.7	19.4	19.1	-2%	0.7
4.0	2015	Cohort	49.6	11.1	13.9	25%	49.6
4.0	2015	NY	1.5	6.5	6.5	1%	1.5
4.0	2015	HILTV	1.5	12.1	9.9	-18%	1.5
4.0	2015	LOWFICO	6.8	13.9	17.6	27%	6.8
4.0	2015	Investor	1.3	11.2	11.5	2%	1.3
4.0	2015	CK	0.2	15.8	35.1	122%	0.2

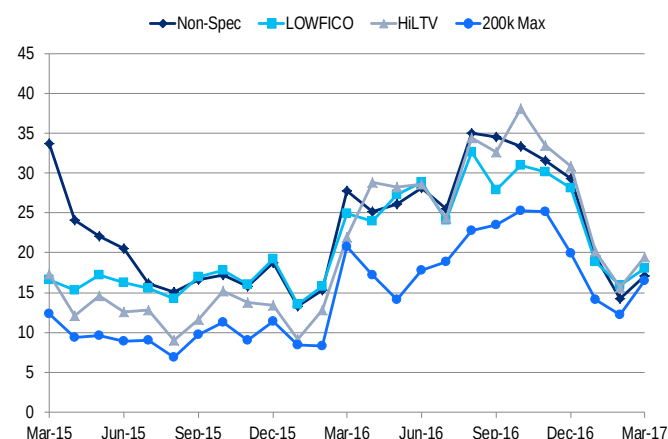
Source: Citi Research, 1010data

Figure 37. NY pools are paying slower than LLB pools in most FN vintages (2015 3.5s, 2014 4.0s)



Source: Citi Research, 1010data

Figure 38. Speeds on low payup stories converged in 2014 FN 4.0s



Source: Citi Research, 1010data

15y and 20y speeds increase on higher daycount and seasonal

FN 15y and FN 20y aggregate speeds increased 25% and 23% in March (Figure 39, Figure 40). As we saw in 30y MBS, Freddie loans saw a much smaller increase. FG 15y and 20y speeds were only up 16% and 13%, respectively. Lower coupon pools saw greater increases, reinforcing the idea that turnover drove the pickup in prepayments. In higher coupons, a large decline in refs likely offset the large increases in daycount and turnover seasonals.

Figure 39. FN 15y Speeds

Coupon	Vintage	Bal (\$bn)	WAC	Feb CPR	Mar CPR	MoM %Chg.
2.0	2013	11	2.61	5.9	8.0	36%
2.0	2012	3	2.63	6.3	8.2	29%
2.5	2015	23	3.05	6.1	7.8	28%
2.5	2014	4	3.15	7.4	9.9	34%
2.5	2013	41	2.93	7.1	9.4	34%
2.5	2012	43	3.00	7.6	9.3	23%
3.0	2015	26	3.48	7.9	9.7	23%
3.0	2014	22	3.53	8.6	11.8	37%
3.0	2013	17	3.51	9.3	11.6	25%
3.0	2012	24	3.45	9.2	11.6	26%
3.0	2011	14	3.45	9.4	12.2	30%
3.5	2015	4	4.04	9.9	12.0	21%
3.5	2014	8	4.05	9.9	13.2	34%
3.5	2013	5	4.02	10.6	13.0	23%
3.5	2012	4	3.97	11.8	12.5	6%
3.5	2011	13	3.91	11.0	13.4	22%
3.5	2010	9	3.92	11.7	13.9	19%
4.0	2011	6	4.38	12.8	14.1	10%
4.0	2010	7	4.42	12.9	15.8	22%
4.0	2009	5	4.48	13.8	15.5	12%
FNM15	ALL	405	3.39	7.9	9.9	25%

Source: Citi Research, eMBS

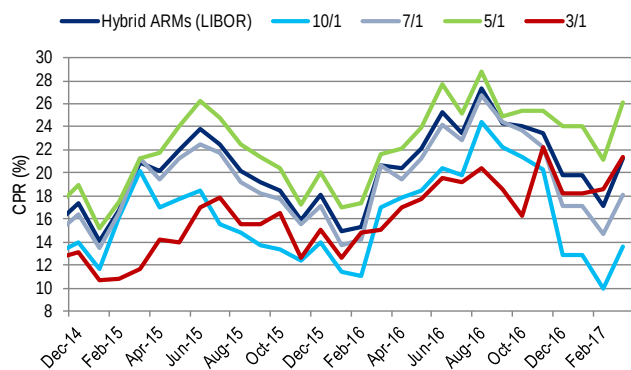
Figure 40. FN 20y Speeds

Coupon	Vintage	Bal (\$bn)	WAC	Jan CPR	Feb CPR	MoM %Chg.
3.0	2015	10	3.76	6.2	8.1	30%
3.0	2014	1	3.86	7.2	10.5	46%
3.0	2013	14	3.58	6.5	8.6	33%
3.0	2012	13	3.58	6.8	8.7	28%
3.5	2015	5	4.15	7.7	11.0	42%
3.5	2014	4	4.25	11.3	12.8	13%
3.5	2013	4	4.11	10.9	11.4	4%
3.5	2012	8	3.99	9.7	10.9	13%
3.5	2011	5	3.99	8.5	10.8	26%
4.0	2014	3	4.65	11.7	14.6	25%
4.0	2011	4	4.48	12.1	13.5	12%
4.0	2010	3	4.45	11.2	13.3	19%
FNM20	ALL	131	3.97	7.3	9.0	23%

Source: Citi Research, eMBS

Aggregate FN hybrid-ARM (LIBOR) speeds increased by 24% to 21.2 CPR (Figure 41) 3/1, 5/1, 7/1 and 10/1 speeds were up 15%, 24%, 23% and 27%, respectively. Vintages near reset like the 2012 5/1s continue to prepay very fast, reaching 51.6 CPR. Within servicers, the more aggressive Tier 3 servicers saw the largest increase in speeds in longer reset 7/1s while March speeds in Tier 1 serviced 5/1s picked up by more than Tier 2 or 3 (Figure 43, Figure 44).

Figure 41. FN Hybrid ARM speed history by sector



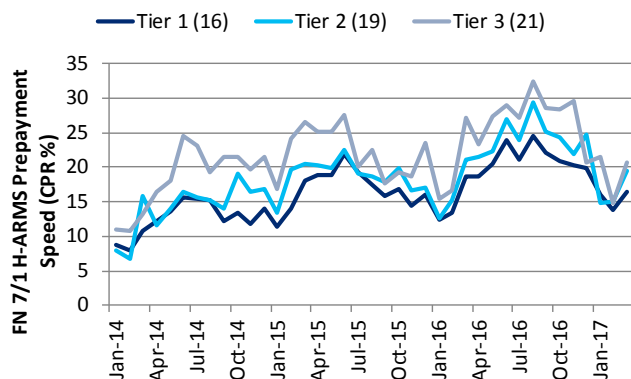
Source: Citi Research, eMBS

Figure 42. FN Hybrid ARM speeds by sector and vintage

ArmType	Vintage	Bal (\$bn)	WAC	Feb CPR	Mar CPR	MoM %Chg.
5/1	2015	3.1	3.11	13.6	20.2	48%
5/1	2014	3.2	3.25	17.1	22.9	34%
5/1	2013	2.3	2.85	20.1	23.9	19%
5/1	2012	2.0	3.05	43.5	51.6	19%
5/1	2011	2.6	3.67	27.9	32.0	15%
5/1	2010	2.1	3.67	22.1	26.4	20%
7/1	2015	4.4	3.34	11.0	14.8	35%
7/1	2014	4.5	3.52	15.3	19.2	25%
7/1	2013	3.6	2.96	12.4	17.8	44%
7/1	2012	4.6	2.96	15.8	19.1	21%
7/1	2011	2.3	3.60	24.4	26.2	7%
7/1	2010	0.9	3.82	30.8	36.2	17%
10/1	2015	0.9	3.56	7.1	10.8	53%
10/1	2014	0.9	3.79	9.7	12.8	32%
10/1	2013	1.5	3.16	10.6	17.7	68%
10/1	2012	1.3	3.26	12.3	14.7	19%
10/1	2011	0.6	3.85	13.4	17.9	34%
10/1	2010	0.2	4.16	18.1	23.6	30%
LIBOR	ALL	60.2	3.36	17.2	21.2	24%

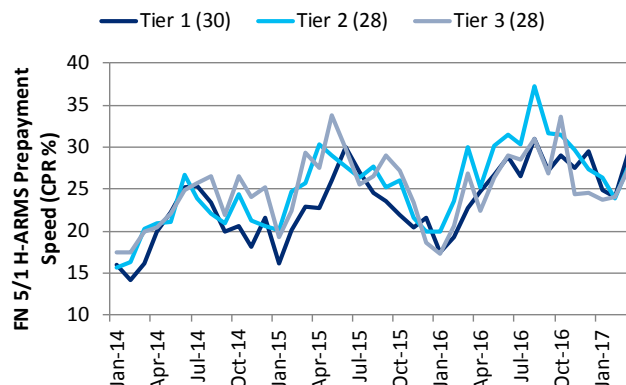
Source: Citi Research, eMBS

Figure 43. FN 7/1 Hybrid-ARM prepayment history by servicers



Note: Tier 1 is defined as Wells, Chase, BoA, US Bank, PNC, 53rd
Tier 2: Ditech, Caliber, Loan Depot, Franklin, Nationstar, Provident, PHH, Pennymac, Stearns
Tier 3: Flagstar, Ocwen, Pingora, Lakeview
Source: Citi Research, 1010data

Figure 44. FN 5/1 Hybrid-ARM prepayment history by servicers



Source: Citi Research, 1010data

Figure 45. 30y aggregate speeds

	Feb CPR	Mar CPR	MoM%
GN2 30y	11.6	13.3	15%
GN1 30y	15.3	17.4	13%
FN 30y	8.8	10.8	22%
GN2-FN	2.8	2.5	
GN1-FN	6.5	6.6	

Source: Citi Research, eMBS

GN2 speeds up 15% narrowing gap with conventionals

GN1 and GN2 aggregate speeds increased by 15% and 3%, respectively, in March (Figure 45). While we expected Ginnie speeds to increase by less than conventional speeds as their higher buyout share makes them less sensitive to turnover seasonals, the absolute level was surprisingly slow as the overall MBS universe prepaid less than we expected.

Figure 46 shows post-HARP GN2 cohort speeds and comparison with FNs. G2 3.0 pools saw smaller percentage increases than their respective FN cohorts although absolute G2-FN speed differentials increased in many cases due to the higher absolute speeds for G2s. In 3.5s, newer vintages saw a smaller increase in speeds in G2s but the older 2012 and 2011 vintages increased more than FNs. Among higher coupons, speeds were more mixed with 4.5s increasing by a larger percentage in G2s and 4.0s and 5.0s increasing by more in FNs.

Figure 46. Post-HARP Ginnie II and G2-FN speeds

Cpn	Vintage	Ginnie II			Speed difference (G2-FN)	
		MoM	Feb CPR	Mar CPR	Feb CPR	Mar CPR
3.0	2015	28%	8.7	11.1	3.7	3.6
3.0	2013	28%	10.1	12.9	4.0	4.4
3.0	2012	26%	10.1	12.7	4.3	4.8
3.5	2015	16%	12.3	14.2	4.8	4.5
3.5	2014	14%	15.2	17.4	5.7	5.7
3.5	2013	19%	13.1	15.7	4.5	4.6
3.5	2012	22%	12.5	15.2	4.0	5.0
3.5	2011	26%	12.6	15.9	4.3	6.4
4.0	2015	6%	23.1	24.5	12.0	10.6
4.0	2014	12%	19.8	22.2	6.7	6.3
4.0	2013	15%	18.5	21.3	5.9	6.5
4.0	2012	13%	15.7	17.8	4.6	4.3
4.0	2011	14%	16.4	18.7	5.5	5.6
4.0	2010	6%	16.0	17.0	4.9	4.4
4.5	2013	10%	21.8	24.0	5.3	6.6
4.5	2011	20%	17.8	21.4	3.8	6.4
4.5	2010	4%	18.5	19.3	3.9	2.9
5.0	2010	7%	20.2	21.7	3.3	2.0
5.0	2009	7%	20.8	22.3	4.0	2.6

Source: Citi Research, eMBS

Figure 47 shows pre-HARP GN1 cohort speeds and comparison with FNs. As we saw in conventionals, speeds were up by less than in post-HARP cohorts. Refi speeds likely remained low in most cohorts as they did in pre-HARP conventional speeds. 2006-2008 GN 6.0s saw larger increases than their corresponding FN cohorts.

Figure 47. Pre-HARP Ginnie I and GN-FN speeds

Cpn	Vintage	Ginnie I			Speed difference (GN-FN)	
		MoM	Jan CPR	Feb CPR	Jan CPR	Feb CPR
5.0	2008	-4%	19.7	19.0	0.9	-2.9
5.0	2005	17%	15.5	18.2	-2.6	-1.7
5.0	2003	10%	14.2	15.6	-1.2	-1.5
5.5	2008	0%	21.3	21.3	1.1	-1.2
5.5	2005	7%	14.3	15.4	-3.1	-4.5
5.5	2004	14%	13.0	14.8	-3.6	-4.7
5.5	2003	9%	14.3	15.6	-1.6	-2.7
6.0	2008	23%	19.7	24.1	0.6	2.2
6.0	2007	11%	17.0	18.9	-4.1	-4.7
6.0	2006	33%	16.7	22.2	-3.8	-1.7

Source: Citi Research, eMBS

The bar chart displays the Federal Reinvestment in billions of dollars from January 2012 to January 2017. The y-axis represents the amount in billions of dollars, ranging from 0 to 50. The x-axis shows time in six-month intervals. The data shows a period of relative stability around \$25 billion from 2012 to 2013, followed by a decline to around \$15 billion in 2014. A significant increase occurs in 2015, peaking at nearly \$40 billion in July, followed by a sharp drop in January 2016. The year 2016 shows a strong recovery, with a peak of approximately \$45 billion in January, before declining again in July 2017.

Month	Fed Reinvestment (\$bn)
Jan-12	25
Jul-12	28
Jan-13	25
Jul-13	35
Jan-14	25
Jul-14	15
Jan-15	20
Jul-15	38
Jan-16	20
Jul-16	30
Jan-17	45
Jul-17	25

Gross Issuance \$93bn, net issuance estimated at \$15bn

Fed purchases to increase to \$23bn in April

We expect Fed reinvestment purchases to increase to \$23bn in the April purchase cycle, up \$5bn from the previous cycle. This increase comes as prepayments increased and \$1.5bn in agency debt is maturing.

Appendix A-1

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